

How bad is the air, and can the data be trusted?

The first piece in this series built on real measurements rather than a simulation: government monitoring stations in Delhi, Mumbai, Kolkata, Chennai, Bengaluru and Hyderabad, from the public OpenAQ archive. There is no answer key this time. There is an archive with a 27-month hole, mislabelled units, a fault code posing as a concentration, frozen sensors and duplicate feeds, each found, counted and handled. Nothing is filled in.

Python, DuckDB, SQL, Excel, Power BI · 107 stations, hourly, 2019-01-01 to 2026-09-30 · CPCB and state boards via OpenAQ

36.1M

valid readings, from
56.0M raw rows

1,089,800

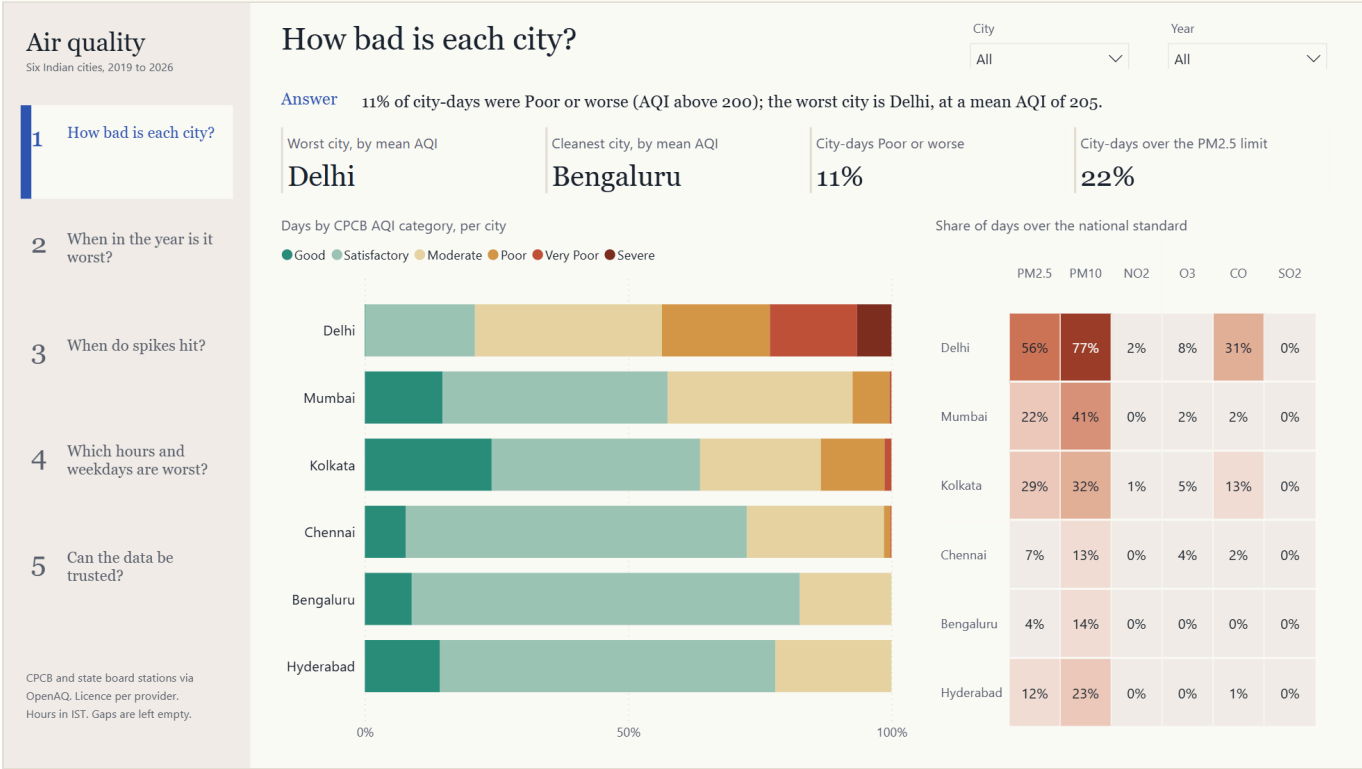
readings set aside, each
under one of 8 named
reasons

27 months

with no data in the
archive, left blank, never
estimated

423 of 423

report numbers matched
against independent SQL



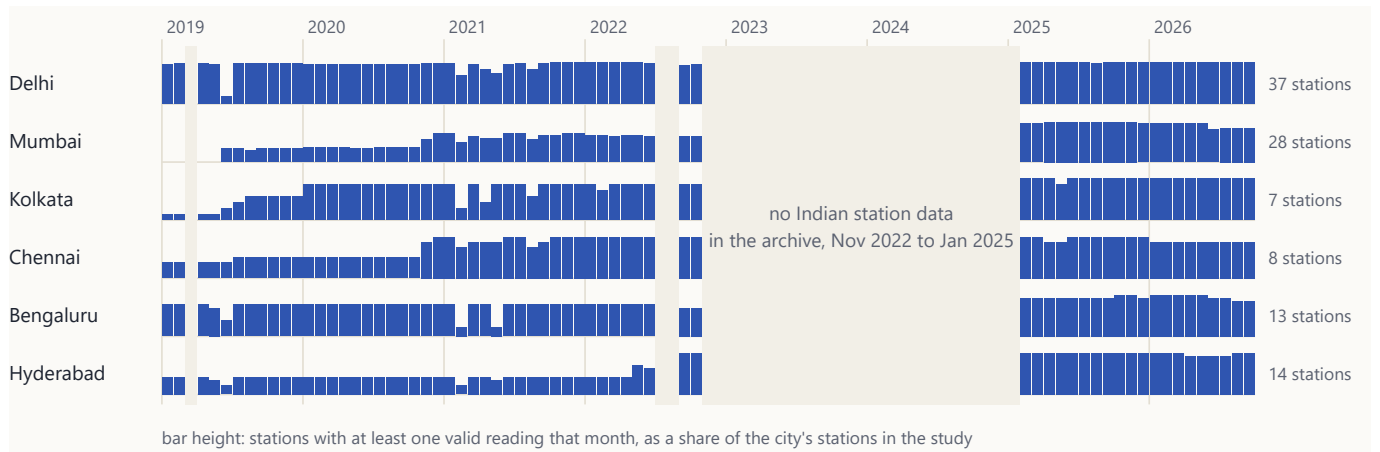
The data

Source	OpenAQ data archive on AWS Open Data, read anonymously over HTTPS: no account, no API key
Underlying data	Continuous monitoring stations of CPCB, state boards, IMD and IITM, published on CPCB's portal
Window	Hourly, on IST, 2019-01-01 to 2026-09-30; valid data from 2019-01-02 17:00 to 2026-09-30 23:00
Stations	107 stations (122 OpenAQ ids): Delhi 37, Mumbai 28, Hyderabad 14, Bengaluru 13, Chennai 8, Kolkata 7
Download	117,993 daily files, 337.4 MB, every file checked against the archive's MD5 and listed with its SHA-256 in a manifest
Rows	56,030,781 raw rows; 37,165,817 readings of the AQI pollutants (PM2.5, PM10, NO2, SO2, O3, CO); 36,075,426 valid; 12,005,342 station-hours; 7,083 city-days with an AQI
Attribution	CPCB and state boards, via OpenAQ; credit both. The licence is set by each provider

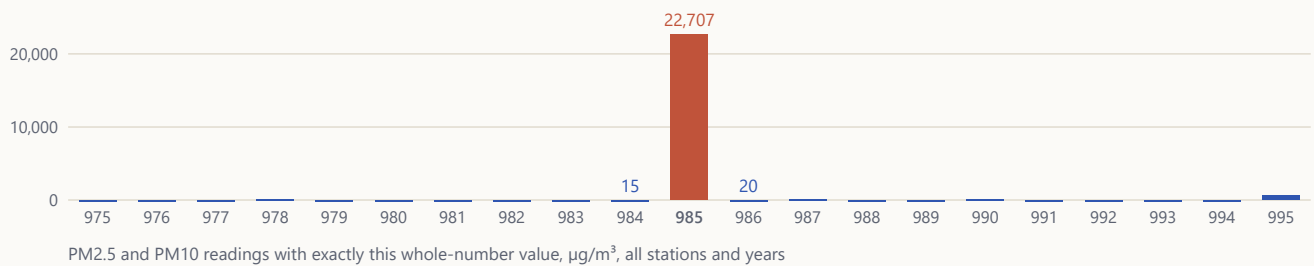
The archive has no station list and OpenAQ's API needs a key, so the stations were found from the archive itself: all 55,965 location ids were listed and the first row of one file per id read with a 2 KB range request. 1,494 fall inside India's bounding box; 170 are government monitors in the six cities, grouped into stations by city, site and agency, each needing 365+ days of files. The download budget is 350 MB.

What real data costs

PROBLEM	WHAT THE DATA SHOWED	WHAT WAS DONE
The archive gap	No Indian station data from 31 Oct 2022 to 19 Feb 2025; 2022 has no July or August data; 185,906 station-days missing in all	Left empty and shown empty. No trend is claimed across the gap
Growing station sets	9 of 28 Mumbai and 3 of 13 Bengaluru stations start in 2025, as does 1 of 7 in Kolkata; 8 of 14 in Hyderabad start in 2022	City averages across years are marked as not like-for-like; station-level tables allow fixed-set comparisons
Mislabelled units	Later files label NO2, SO2 and CO as "ppb". CO "ppb" has a median of 0.66, where CO labelled $\mu\text{g}/\text{m}^3$ has 710 (0.71 mg/m^3); NO2 "ppb" has a median of 17.4, against 17.6 $\mu\text{g}/\text{m}^3$ at the same stations earlier	The values are CPCB's own units under a wrong label: 12,171,979 readings at 106 stations relabelled, not converted. A check retests the evidence on every build
Fault code 985	Exactly 985 appears 22,707 times in PM readings, against 15 for 984 and 20 for 986, at 21 stations	Set aside as a status code, not a concentration
Frozen values	5,890 runs of an identical value for 8 or more consecutive hours; the longest is 446 hours of NO2 at IDA Pashamylaram (Hyderabad)	414,505 readings set aside
Duplicate feeds	At 2 stations an old and a new OpenAQ id both publish from February 2025. Pusa (Delhi): the same value at 31 of 18,368 shared timestamps, correlation 0.76; Worli (Mumbai): the same value at 3 of 29,630 shared timestamps, correlation 0.19	Two different instruments, not one feed twice: 475,812 conflicting readings set aside, with no basis to choose. Identical copies keep one
Moving coordinates	8 stations change their published coordinates by over 100 m; the largest shift is 363 km, a geocoding error	Kept as one station, latest coordinates used, counted



The archive's last Indian reading before the hole is on 31 Oct 2022; the next is on 19 Feb 2025: 841 days, 30% of the window. The bars also show station sets growing: 9 of 28 Mumbai stations first report in 2025, and 8 of 14 Hyderabad stations in 2022.



A real concentration does not prefer one integer. 985 is an analyser status code that reaches the archive as a value; left in, it would add thousands of hours of Severe air that never happened.

Every reading gets one disposition

Each reading of the six AQI pollutants receives exactly one disposition, valid or a named issue, and only valid readings reach the facts. 17 kinds of issue are logged across readings, hours, days and stations, each with its action and count (the report's trust page, below). 8 of them set readings aside, 1,089,800 in all. The rest keep the reading and count it: 4,183 particulate readings sit at the analyser's ceiling of 1000 $\mu\text{g}/\text{m}^3$, where the true value may be higher, and 198,026 station-days have fewer than 16 valid hours and get no daily value or AQI.

The pipeline

Discovery and download are separate, resumable scripts; the download refuses a plan above its budget and writes a manifest with every file's URL, size, MD5 and SHA-256. Staging types every reading, converts timestamps to IST (readings are period-ending: a 17:00 stamp belongs to the 16:00 hour), applies the unit rules and gives each reading its one disposition. The model layer builds the dimensions and hourly and daily facts; a day needs 16 valid hours. The marts score CPCB's National AQI (24-hour means for particulates, NO₂ and SO₂, the highest 8-hour mean for O₃ and CO; three pollutants including PM_{2.5} or PM₁₀ for a station AQI; the city AQI as the mean of its stations), then months, seasons, exceedance of the national standards, spikes, hourly profiles and completeness.

The export to Excel and Power BI is blocked if any check fails. The workbook (12 sheets, 9 Excel Tables, month by city matrices coloured by AQI category) and the Power BI project are both generated from the same warehouse. A full build from the raw files took about 26 minutes.

Proving it

- **23 of 23 reconciliation checks pass** on every build: every raw row staged once, one disposition per reading, hourly and daily values recomputed from their readings, no hour or day invented, AQI sub-indices against an independent implementation of CPCB's breakpoints, city and month figures recounted, spike episodes rebuilt, data-quality counts reconciled and the unit evidence retested.
- **Mutation test, 23 of 23 caught.** A copy of the warehouse is broken once per check (a duplicated reading, an invented hour, a wrong sub-index, a spike run cut short...). Each break must fail its check, or the check is not proving anything.
- **423 of 423 report numbers match.** The report is opened in its own Power BI window; 56 of 56 measures evaluate without error, and every figure on the 5 pages (cards, answer sentences rebuilt word for word, heatmap cells, the spike table, the issue log) is compared with independent SQL.
- **Nothing clipped.** 305 of 305 single lines and 36 of 36 wrapped texts are measured with the report's real fonts against their boxes, under every slicer combination; 14 of 14 grids fit without scrollbars.
- **Byte-identical rebuilds.** Two builds from the raw files give the same 18 of 18 output files (every exported table and the workbook); a regenerated Power BI project matches the committed files.

The report

Five pages, generated as a Power BI project (a TMDL model of 14 tables, 20 relationships and 56 measures, and a PBIR report) by `powerbi/build_pbip.py`. The layout is a question rail: one plain question per page down the left, and each page opens with its one-sentence answer, computed by a measure, before any chart. Colours follow CPCB's AQI categories; nothing in a sentence on the report is typed by hand.

Air quality

Six Indian cities, 2019 to 2026

1

How bad is each city?

2

When in the year is it worst?

3

When do spikes hit?

4

Which hours and weekdays are worst?

5

Can the data be trusted?

CPCB and state board stations via OpenAQ. Licence per provider. Hours in IST. Gaps are left empty.

Can the data be trusted?

Answer28% of possible station-hours hold a valid reading; 1,089,800 readings were set aside, each under a named reason.

Stations

Valid station-hours

Readings set aside

Station-days missing

Completeness

107

12,005,342

1,089,800

185,906

28%

Completeness: valid station-hours out of possible, by city and year

20192020202120222023202420252026

Delhi

38%48%27%17%0%0%76%82%

Mumbai

12%17%17%10%0%0%61%60%

Kolkata

21%44%23%17%0%0%76%82%

Chennai

15%25%24%19%0%0%66%66%

Bengaluru

26%32%16%11%0%0%60%58%

Hyderabad

18%24%10%11%0%0%71%75%

Issue log with the action taken, biggest first

Issue

Action taken

Count

gas unit labelled ppb (value is CPCB µg/m³, CO mg/m³)

kept, label corrected

12,171,979

conflicting duplicate

set aside

475,812

stuck value

set aside

414,505

incomplete day (under 16 valid hours)

no daily value or AQI

198,026

day missing from the archive

no data

185,906

zero particulate reading

set aside

112,703

negative value

set aside

46,713

PM2.5 above PM10 in the same hour

kept, reported

26,955

instrument fault code 985

set aside

22,707

above plausible maximum

set aside

16,926

particulate at the analyser ceiling (999.99-1000 µg/m³)

kept, reported (true value may be higher)

4,183

isolated one-hour particulate spike

kept, reported

1,549

duplicate across station ids

one copy kept

591

unknown unit

set aside

363

unparseable value

set aside

71

station published under several OpenAQ ids

ids merged into one station

15

published coordinates changed by over 100 m

kept, one station; latest coordinates used

8

Can the data be trusted? 73% of possible station-hours hold a valid reading in 2025-2026; the empty columns are the archive gap.

Air quality

Six Indian cities, 2019 to 2026

1 How bad is each city?

2 When in the year is it worst?

3 When do spikes hit?

4 Which hours and weekdays are worst?

5 Can the data be trusted?

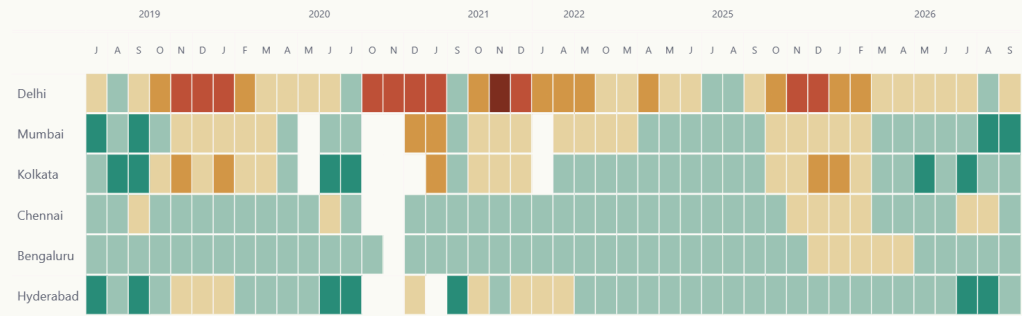
CPCB and state board stations via OpenAQ. Licence per provider. Hours in IST. Gaps are left empty.

When in the year is it worst?

City Year

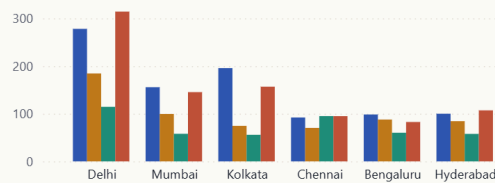
Answer Winter (Jan-Feb) is the worst season, at a mean AQI of 155, against 74 in the monsoon (Jun-Sep).

Mean AQI by city and month, coloured by CPCB category (months with 15+ days; the archive has no data Nov 2022 to Feb 2025)



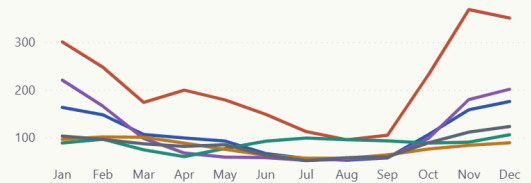
Mean AQI by season, per city

Winter Pre-monsoon Monsoon Post-monsoon



Typical year: mean AQI by calendar month

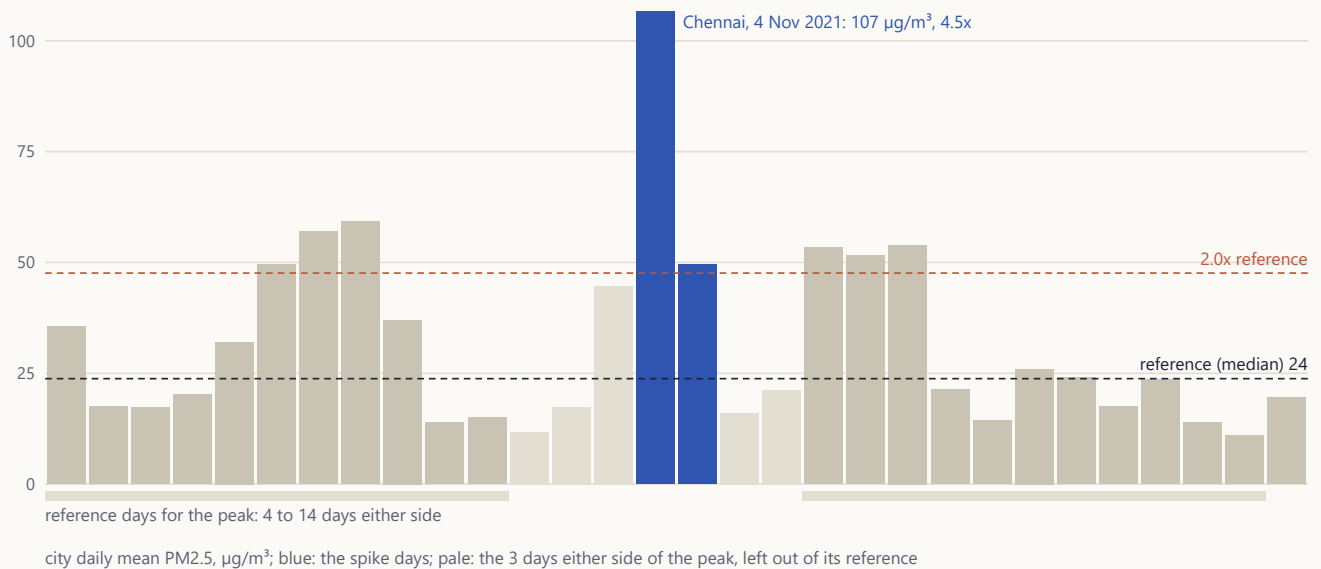
Delhi Mumbai Kolkata Chennai Bengaluru Hyderabad



When in the year is it worst? Months with under 15 days of AQI stay blank; the archive gap is not drawn as clean air.

Short spikes, found from the data alone

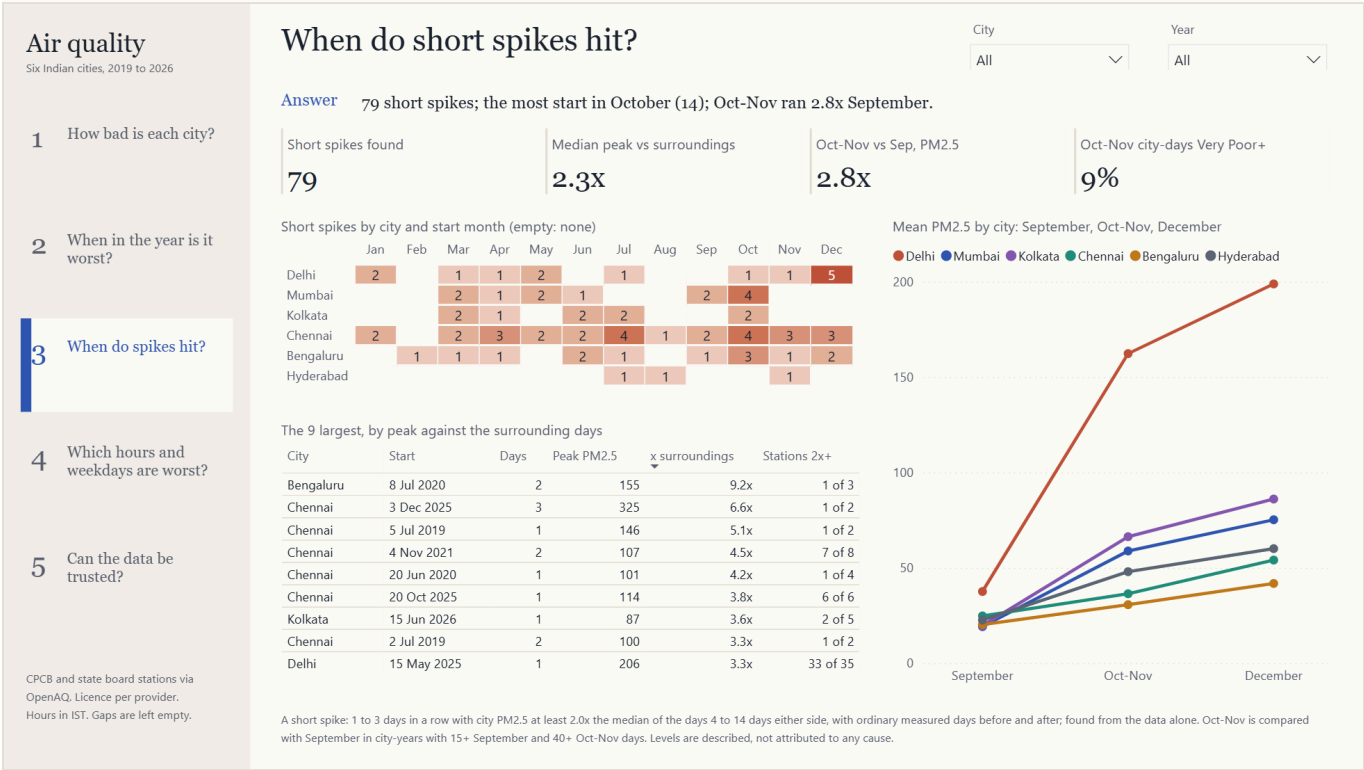
There is no event calendar, no labels and no cause: a spike is defined only by how a day compares with the days around it.



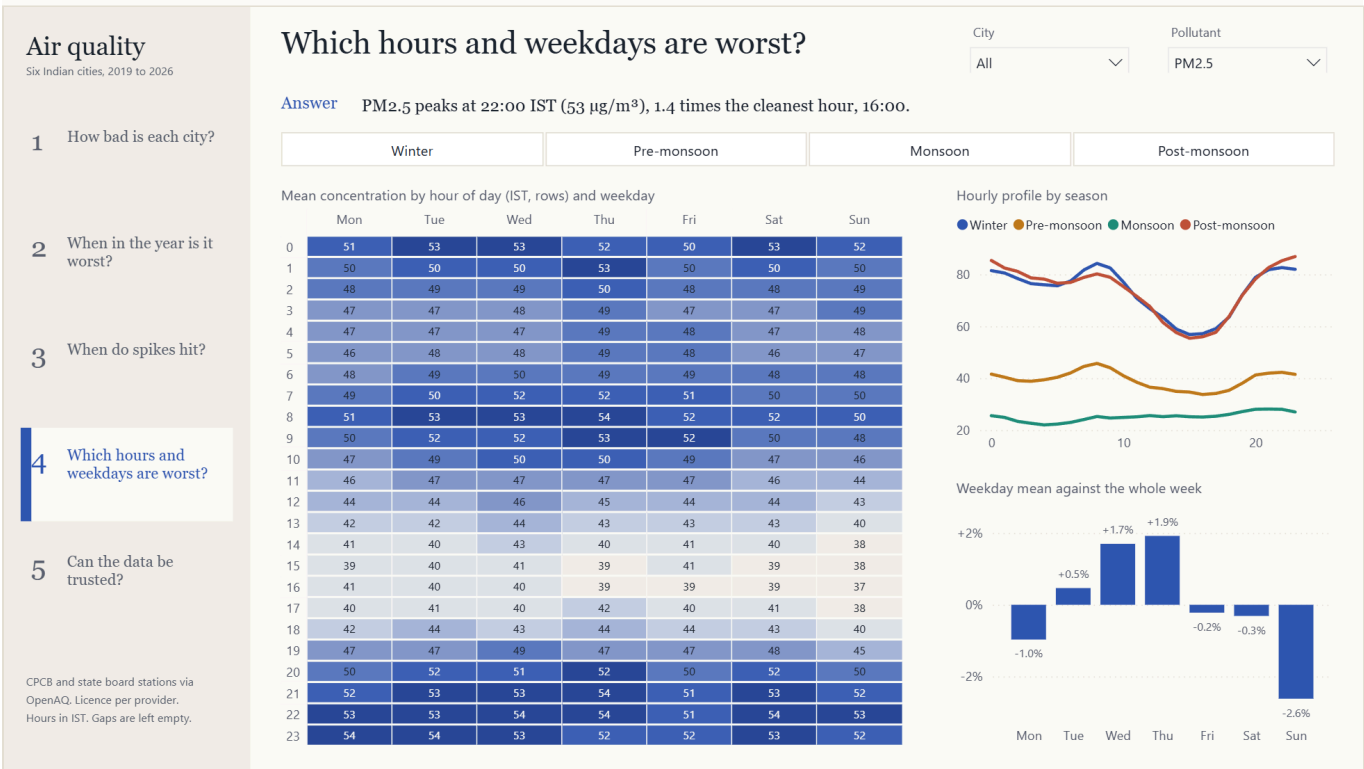
The largest spike carried by most of its city's stations: Chennai, 2 days from 4 Nov 2021, peak 107 $\mu\text{g}/\text{m}^3$ against a reference of 24 (4.5x), with 7 of 8 stations at 2.0x the city reference that day. The reference is computed for every day; the lines show the peak day's.

For each city-day, the reference is the median city PM2.5 of the days 4 to 14 days before and after it. The 3 days either side are left out, so an episode of up to 3 days never sets its own reference, and a day is evaluated only when 6 of the 11 reference days are measured on each side. A spike day is at least 2.0x its reference: 144 of 5,724 evaluated city-days (2.5%). A short spike is a run of 1 to 3 such days with measured ordinary days on both sides.

79 episodes: Chennai 28, Delhi 14, Bengaluru 13, Mumbai 12, Kolkata 9, Hyderabad 3. Per 100 evaluated city-days they start most often in October (3.5; 14 of 405 days) and least often in February (0.3). A city value is a mean of stations, so each episode records how many stations carried it: in 22 episodes a single station of several carries the peak day, and 2 rest on a day with only one station reporting. They are shown as such, not hidden. Not counted: 3 runs longer than 3 days and 18 runs next to a day without a reading or reference, where the length cannot be known.



When do spikes hit? October-November is compared with September only in city-years with enough days of both.



Which hours and weekdays are worst? Weekday differences are a few percent.

What the data says

401

Mean AQI in Delhi, Nov 2021, the worst city-month: 30 of 30 days Very Poor or Severe. Next is Delhi Nov 2025 (386, 28 of 29 days Very Poor or Severe). The worst month outside Delhi is Kolkata Jan 2021 at 271.

56.5%

Of measured days in Delhi have city PM2.5 above the national 24-hour standard of 60 $\mu\text{g}/\text{m}^3$. Every city crosses it on some days: Kolkata 29.4% (336 of 1,141), Mumbai 22.1% (256 of 1,158), Hyderabad 11.8% (139 of 1,180), Chennai 6.8% (81 of 1,185), Bengaluru 3.6% (42 of 1,178).

3.3x to 5.3x

Delhi's October-November PM2.5 against its September, in years with 15+ September and 40+ October-November days: 2019: 41 to 138 $\mu\text{g}/\text{m}^3$ (3.3x); 2021: 35 to 185 $\mu\text{g}/\text{m}^3$ (5.3x); 2025: 37 to 164 $\mu\text{g}/\text{m}^3$ (4.5x). Across all cities, 9% of October-November city-days are Very Poor or worse. These are levels only: the dataset has no emissions or weather data, so no cause is attributed.

22:00

The worst hour for PM2.5 across all cities, at 1.4x the cleanest. In 4 of the six cities the worst hour falls in the evening or around midnight; Chennai (08:00) and Mumbai (08:00) peak in the morning. Worst against cleanest hour, $\mu\text{g}/\text{m}^3$: Delhi 23:00 (111) against 16:00 (65), 1.71x; Kolkata 00:00 (55) against 15:00 (38), 1.42x; Bengaluru 20:00 (36) against 04:00 (27), 1.36x; Hyderabad 20:00 (42) against 14:00 (32), 1.30x; Chennai 08:00 (37) against 02:00 (29), 1.29x; Mumbai 08:00 (45) against 18:00 (36), 1.25x.

2.3x

Hyderabad's mean daily PM2.5 in its worst season (post-monsoon, 51.1 $\mu\text{g}/\text{m}^3$) against its cleanest (monsoon, 22.3 $\mu\text{g}/\text{m}^3$).

Limits and next steps

- The 27-month gap splits the record in two. Nothing can be said about how the air changed between 31 Oct 2022 and 19 Feb 2025.
- Station sets grow, so city averages across years are not like-for-like. A fixed panel of stations present in every year would be the honest way to compare years.
- There is no weather or emissions data, so the analysis describes levels and timing, never causes. Joining IMD weather would be the next step towards explaining them.
- CPCB's open-ended Severe band is closed at its lower bound plus the width of the band below; NH3 and lead are not in the archive. Power BI has no free public link, so the report ships as a project with these images.

Every figure and chart is computed from the warehouse and the build logs by docs/render_docs.py and docs/charts.py. Data: CPCB and state boards, via OpenAQ.